How Regulated Systems Produce Responsibility: A Four-Layer Framework for Interaction Design

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Abstract Regulated environments — pharmaceutical manufacturing, clinical healthcare, aviation, financial compliance — are typically evaluated as documentation systems: whether records are accurate, procedures are followed, and audit trails are complete. This paper introduces the **Regulatory HCI** framework, which reframes regulated environments as interaction systems and establishes a generalizable mechanism by which system design produces accountability conditions under compliance constraints. The framework identifies four interdependent layers at which design choices determine accountability conditions: an accessibility layer, which shapes what information practitioners can reach; a structural layer, which shapes how document relationships are visible and navigable; a coordination layer, which shapes how work is distributed and observable across actors; and a reasoning layer, which shapes how explanations are generated, compared, and stabilized. Grounded in four prototype-based empirical studies conducted in pharmaceutical GMP environments (each examining one layer, each with $N = 20$ GMP-experienced participants), the framework establishes a consistent mechanism across all layers: system design creates conditions that are invisible to formal quality indicators, that redistribute labor unequally across a workforce stratified by experience, and that attribute the resulting burden to individuals rather than to the structural conditions that produced it. The framework extends existing sociotechnical systems theory and value-sensitive design into compliance-constrained environments, offering an account of how responsibility is produced through system structure rather than assigned after the fact.

Keywords: Regulatory HCI, accountability, compliance-constrained design, sociotechnical systems, GMP, judgment preservation, invisible work, interaction design

Introduction

Regulated environments — pharmaceutical manufacturing, clinical healthcare, aviation, financial compliance — are typically evaluated as documentation systems: whether records are accurate, whether procedures are followed, whether audit trails can be reconstructed. This framing treats regulation as a set of requirements imposed on work, and treats system design as the problem of satisfying those requirements efficiently. It asks whether users perform tasks correctly. It does

not ask how system design determines which tasks are available to perform, which interpretations remain open, or which coordination work gets recognized.

I argue that regulated environments should be understood as interaction systems — systems that actively shape how users access information, interpret it, coordinate with others, and ultimately produce accountable decisions. The design of these systems is not neutral with respect to accountability: it determines which explanations remain viable, which pathways are visible, and which forms of labor are measured. Responsibility, in this framing, is not only assigned after the fact. It is produced through the structure of the system itself.

I introduce the **Regulatory HCI** framework for analyzing this production, and derive three design principles that translate it into actionable guidance. Regulatory HCI examines how system design structures the conditions under which accountability emerges, treating compliance-constrained interaction as a design problem rather than a compliance problem. The framework identifies four interdependent layers at which system design shapes accountability conditions: an accessibility layer, which determines how users reach information; a structural layer, which determines how document relationships are represented and navigable; a coordination layer, which determines how work is distributed and made visible across actors; and a reasoning layer, which determines how explanations are generated, compared, and stabilized (Figure 1).

The framework is grounded in four empirical studies conducted in pharmaceutical GMP environments, each examining one layer through prototype-based evaluation with $N = 20$ GMP-experienced participants. Full accounts of each study are reported in companion papers currently under review: the accessibility layer study (Kim, 2026a), the structural layer study (Kim, 2026b), the coordination layer study (Kim, 2026c), and the reasoning layer study (Kim, 2026d). This paper synthesizes across the four studies to establish the cross-layer mechanism, introduce the design principles that follow from it, and develop the theoretical argument that regulated environments are interaction systems in which responsibility is produced rather than assigned.

This paper makes three contributions. First, it introduces Regulatory HCI as a theoretical framework that reconstitutes regulated environments as interaction systems, shifting the design question from compliance optimization to accountability production. Second, it establishes a four-layer model linking system design to accountability production across access, structure, coordination, and reasoning. Third, it derives three design principles — upstream stabilization, observability over optimization, and judgment preservation — that translate the framework into actionable guidance for designers working in compliance-constrained environments.

Related Work

Sociotechnical Systems and the Design of Work

The premise of the Regulatory HCI framework — that system design shapes the conditions under which accountability emerges — draws on a long tradition of sociotechnical systems research. Baxter and Sommerville (2011) establish that the design of work systems cannot be separated from the social conditions in which they operate: technical design choices produce organizational outcomes, and organizational conditions shape how technical systems are used. Regulatory HCI extends this argument into compliance-constrained environments specifically — environments in which the sociotechnical stakes of design are particularly high because accountability is formally required and legally consequential.

Values in Design

Friedman et al.'s (2002) Value Sensitive Design provides a complementary theoretical foundation. VSD establishes that technology design is never value-neutral: design choices embed assumptions about who matters, what counts as success, and how competing interests are balanced. Regulatory HCI applies this insight to accountability specifically: in regulated environments, design choices embed assumptions about how responsibility is distributed, which forms of labor are recognized, and whose epistemic access to information is treated as a design concern.

Documents as Organizational Action

Harper's (1998) ethnographic account of documentation practices at the IMF establishes the foundational premise of the framework's structural layer: documents are not neutral containers of information but active participants in organizational action. Their accessibility, their linkage to other documents, and the navigational demands they impose on users are constitutive of organizational practice. When document relationships are visible and traversable, practitioners can reconstruct accountable pathways; when they are not, that reconstruction work falls to individuals.

Awareness and the Visibility of Work

Dourish and Bellotti's (1992) account of awareness in collaborative work systems establishes the theoretical basis for the framework's coordination layer. They show that awareness — understanding of the activities of others that provides context for one's own action — is critical

to successful coordination and is poorly served by mechanisms that require explicit information-sharing. The framework's coordination layer extends this insight: it treats the invisibility of review friction not as a communication failure but as a design failure.

Invisible Work and the Attribution of Labor

Star and Strauss (1999) establish that work performed in the gaps of formal systems tends to become invisible — not captured by indicators, not attributed to the conditions that produced it, and not recognized as labor by the organizations that depend on it. This insight runs through all four layers of the Regulatory HCI framework. At the accessibility layer, retrieval burden is invisible upstream of formal errors. At the structural layer, traceability reconstruction is invisible in quality metrics. At the coordination layer, review friction is invisible in closure records. At the reasoning layer, comparison collapse is invisible in final investigation documents.

Situated Action and the Conditions of Accountable Practice

Suchman's (1987) account of situated action provides the theoretical spine of the framework. Suchman establishes that action is not plan-driven but shaped by the immediate conditions of its execution — what is visible, accessible, and interpretable at the moment a decision must be made. Regulatory HCI applies this principle systematically across four layers: at each layer, the conditions of action shape the outcomes of action, and those conditions are themselves shaped by system design.

Human-AI Interaction and the Preservation of Judgment

The framework's reasoning layer draws on Amershi et al.'s (2019) guidelines for human-AI interaction, which establish that AI systems should support human control, make their limitations transparent, and avoid pre-empting human judgment. The framework's judgment preservation principle extends this guideline into a design constraint: in compliance-constrained environments, the goal of AI support is not to accelerate decisions but to preserve the conditions under which genuine human judgment is possible.

Epistemic Access and the Distribution of Knowledge

Fricker's (2007) account of epistemic injustice provides theoretical grounding for the framework's unifying theme. Fricker establishes that epistemic access — the ability to know, to be heard as a knower, and to participate in the production of knowledge — is not equally distributed, and that structural conditions determine who can contribute to and benefit from shared knowledge practices. Applied to regulated work systems, design choices which restrict

information access, obscure document relationships, or suppress comparison are not merely usability failures: they are conditions that determine whose knowledge counts and who bears the cost of structural invisibility.

AI-Assisted Decision Making and Judgment-Preserving Design

Within HCI, work on AI-assisted decision making has established that interface structure actively shapes which information becomes salient to practitioners navigating complex regulated environments. Wang and Yin (2021) demonstrate that explanation interfaces alter how decision makers engage with outputs depending on domain expertise. Green and Chen (2019) show that algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible, grounding the framework's judgment preservation principle.

The Regulatory HCI Framework

Overview

Regulatory HCI is a framework for analyzing how system design shapes accountability conditions in compliance-constrained environments. It identifies four interdependent layers at which design choices determine the conditions under which accountable action is possible: an accessibility layer governing information reach, a structural layer governing document relationship visibility, a coordination layer governing the observability of collaborative work, and a reasoning layer governing how explanations are generated and stabilized (Figure 1).

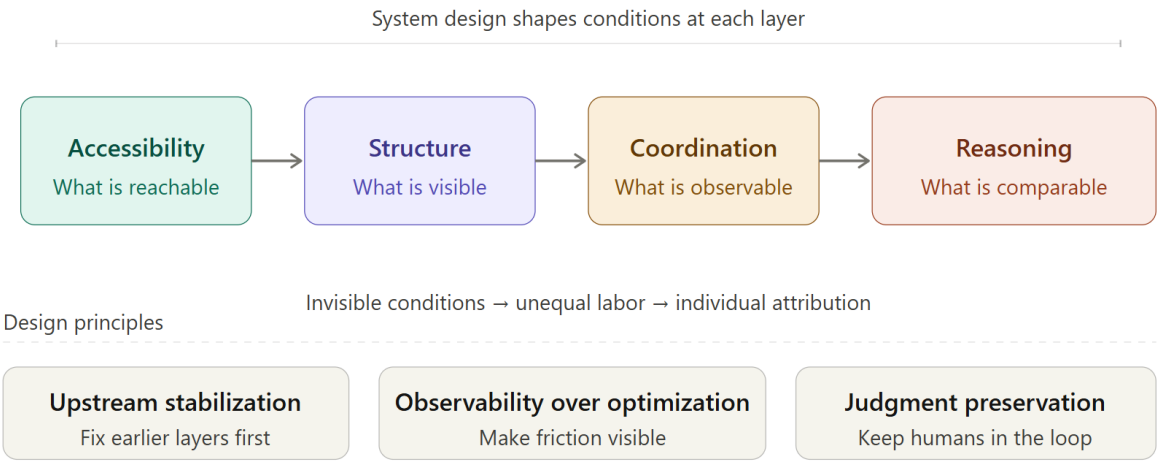


Figure 1: The Regulatory HCI framework: four interdependent layers at which system design shapes accountability conditions in compliance-constrained environments. Arrows indicate the direction of causal dependency: conditions at earlier layers shape the conditions available at later ones.

The framework is grounded in a consistent theoretical commitment: following Suchman (1987), it treats action as situated — shaped by the immediate conditions of its execution rather than by formal plans or procedures alone.

Layer 1: Accessibility

The accessibility layer concerns how practitioners reach information — the conditions under which documents, procedures, and data are locatable, retrievable, and ready-to-hand at the moment of action. When access is constrained, interpretation begins from incomplete information. A practitioner who cannot locate the most applicable procedure makes decisions under conditions that the system has structured. The resulting interpretive variability is upstream of any formal error.

Design implication: accessibility interventions should operate externally, over metadata or retrieval infrastructure, without modifying validated source documents.

Layer 2: Structure

The structural layer concerns how document relationships are represented — the conditions under which connections between documents are visible, navigable, and reconstructable. When linkage structure is incomplete, the work of reconstructing accountable pathways is performed

by individuals, not by systems, and distributed unequally across a workforce stratified by experience.

Design implication: structural interventions should make linkage relationships navigable without modifying validated source files.

Layer 3: Coordination

The coordination layer concerns how work is distributed and made visible across actors. When the conditions of review are not visible, accountability for delay is assigned without access to the coordination conditions that produced it.

Design implication: coordination interventions should surface workflow patterns as interpretable signals without modifying validated documentation systems or automating quality decisions.

Layer 4: Reasoning

The reasoning layer concerns how explanations are generated, compared, and stabilized. Under closure pressure, comparison between alternative explanations collapses before reasoning has fully developed. The resulting causal account reflects the conditions of its production rather than systematic comparison of alternatives.

Design implication: reasoning interventions should preserve comparison between alternatives. AI support in this layer should function as a comparison enforcer rather than an explanation source.

The Mechanism: From Design to Accountability

Across all four layers, a consistent mechanism operates. System design shapes the conditions of action at each layer; the conditions at each layer shape the conditions at the next; and the cumulative effect of design choices across all four layers determines how responsibility is produced and distributed. In each case, design choices that leave conditions invisible, inaccessible, or unsupported redistribute labor and accountability to individuals — while the conditions that produced the redistribution remain invisible in formal records.

Empirical Program

The Regulatory HCI framework is grounded in four prototype-based studies conducted in pharmaceutical GMP environments. Each study examines one layer of the framework, using a combination of synthetic document corpora, behavioral observation, and pilot evaluation with $N = 20$ GMP-experienced participants.

Layer 1: Accessibility (Kim, 2026a)

The central finding was not that a retrieval tool reduces search time — though it does, from approximately six minutes to approximately one minute — but that access conditions shape the behavioral repertoire practitioners allow themselves. When SOP retrieval is structurally difficult, practitioners act within a personally known subset of the procedural archive rather than those most applicable. 18 of 20 participants rated the prototype as intuitive; all 20 expressed adoption intent.

Layer 2: Structure (Kim, 2026b)

The traceability visibility prototype was associated with a 30–60% reduction in path reconstruction time; all 20 participants completed target pathway reconstruction, rated ease of use at 5.0/5, and expressed adoption intent. Loopback behavior was observed across all 20 sessions, with less experienced participants averaging 7–10 loopbacks versus 2–3 for experienced practitioners — not a reasoning gap but a design gap.

Layer 3: Coordination (Kim, 2026c)

All 20 participants rated both dashboard components 5.0/5 on usability and expressed adoption intent. The central finding was that all 20 participants anticipated reduced accountability asymmetry from friction visibility — describing the dashboard as infrastructure for making coordination asymmetry discussable rather than invisible.

Layer 4: Reasoning (Kim, 2026d)

15 of 20 participants produced single-cause, person-centered explanations at the initial timepoint, with confidence ratings averaging 1 out of 5 — indicating that early convergence reflects institutional pressure rather than epistemic certainty. With AI scaffolding, 18 of 20 produced broader causal accounts; 12 of 15 shifted from person-centered to systemic explanations. All 20 expressed adoption intent.

Cross-Layer Synthesis

Across all four studies, a consistent structural pattern emerges: system design creates conditions invisible to formal quality indicators; the invisible condition produces labor absorbed by individuals, distributed unequally across experience levels, and attributed to individual failure rather than structural design; and prototype intervention demonstrates that making the invisible

condition visible reduces the labor burden without modifying validated systems or displacing human judgment.

Design Principles

Principle 1: Upstream Stabilization

Improving accountability outcomes requires stabilizing conditions at earlier layers before intervening at later ones. Retrieval system design is not a secondary concern — it is a precondition for structural, coordination, and reasoning interventions to function.

Principle 2: Observability over Optimization

The design goal in compliance-constrained environments is not efficiency but visibility. Regulatory HCI optimizes for observability: what is currently invisible in the system, which labor is unattributed, which patterns are distributed without a shared representation.

Principle 3: Judgment Preservation

In regulated environments, any system intervention that automates quality decisions or pre-empts human accountability introduces regulatory risk. Judgment preservation constrains what AI can do: at each layer, computational support improves the conditions under which human judgment operates without substituting for it.

Discussion

Regulatory HCI as a Reframing

The dominant framing treats regulation as a constraint on design. Regulatory HCI inverts this framing: it treats regulation as the context in which accountability is produced, and treats design as a participant in that production. The question is not how to design systems that satisfy regulatory requirements; it is how to design systems that shape accountability conditions in ways that are fair, observable, and supportive of genuine human judgment.

Epistemic Access as a Unifying Theme

Across all four layers, the empirical studies converge on a finding the framework makes explicit: system design determines epistemic access — who can see what information, navigate which pathways, observe which patterns, and sustain which comparisons. The practitioners who bear

the greatest burden are consistently those with the least accumulated implicit knowledge of the system. This asymmetry is not a training gap; it is a design gap. Fricker's (2007) account of epistemic injustice is directly applicable: structural conditions restrict epistemic access not by competence but by accumulated implicit knowledge the system itself does not provide.

The Limits of Interface Design

The organizational adoption barriers identified across the empirical studies point to a limitation that interface design alone cannot address. Transparency — the core value the Regulatory HCI tools instantiate — may not be organizationally incentivized. Addressing those structures requires engagement with quality culture, audit standards, and institutional accountability — work that is organizational and regulatory as much as it is design work.

Generalizability

The framework was developed in pharmaceutical GMP environments, but the mechanism it identifies is not domain-specific. Any regulated environment in which accountability is formally required and system design is consequential — clinical healthcare, aviation maintenance, financial compliance, legal practice — is a candidate for Regulatory HCI analysis.

Conclusion

This paper introduced the Regulatory HCI framework for explaining how regulated systems produce responsibility through interaction conditions. The four-layer model — accessibility, structure, coordination, and reasoning — identifies a consistent mechanism: system design creates conditions invisible to formal quality indicators, redistributes labor unequally across experience levels, and attributes the resulting burden to individuals rather than to structural conditions.

Three design principles follow: upstream stabilization holds that accountability conditions must be addressed at earlier layers before interventions at later ones can be effective; observability over optimization holds that the design goal is visibility rather than efficiency; and judgment preservation holds that computational support improves the conditions under which human judgment operates rather than substituting for it.

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